WORKSHOP INTRODUCTION

Network Programming with Java

From Network Concepts to Socket Programming

Foundations · Protocols · TCP/UDP · Java Sockets · Lab Exercises

Workshop Agenda

What we will cover today

01	**Network Foundations** Concepts, Types & Topologies	02	**Network Layers** OSI Model & TCP/IP Stack
03	**Network Devices** Hardware, NIC & MAC Addresses	04	**IP Addressing** IPv4, IPv6, Subnetting & NAT
05	**TCP vs UDP** Reliability, Speed & Use Cases	06	**Java Sockets** Programming the Network in Java

MODULE 01

Network Foundations

Understanding what a network is, its types, and topologies

What is a Computer Network?

Definition
A collection of interconnected devices that share resources and communicate using agreed-upon protocols.

LAN
Local Area Network
Building / Floor

MAN
Metropolitan Area Network
City / Campus

WAN
Wide Area Network
Country / Globe

PAN
Personal Area Network
< 10 meters

Architecture Models

Client-Server

Centralized model. Clients request services; servers provide them. Example: Web browsing, Email.

Peer-to-Peer (P2P)

Decentralized model. Each node acts as both client and server. Example: BitTorrent, Blockchain.

MODULE 02

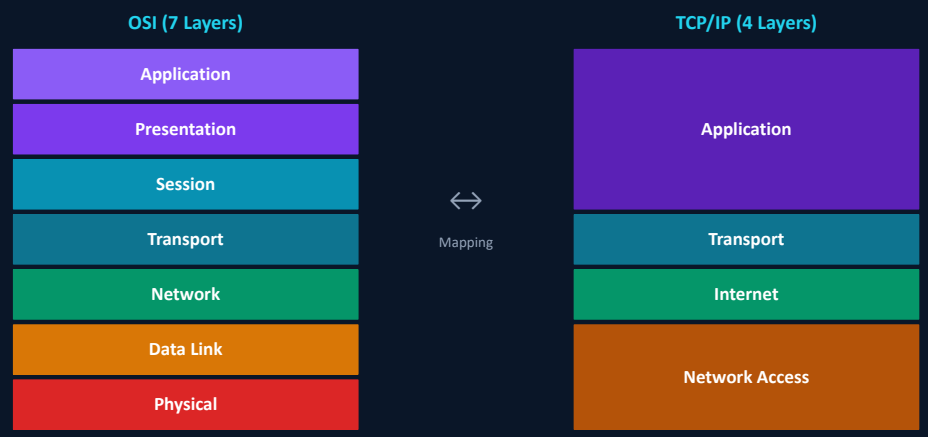
Network Layers

OSI Model & TCP/IP Stack — How data travels across the internet

The 7-Layer OSI Model

Layer	Name	PDU	Protocols / Examples
7	Application	Data	HTTP, FTP, DNS, SMTP
6	Presentation	Data	SSL/TLS, JPEG, ASCII
5	Session	Data	NetBIOS, RPC, PPTP
4	Transport	Segment	TCP, UDP, port numbers
3	Network	Packet	IP, ICMP, routers
2	Data Link	Frame	Ethernet, MAC, switches
1	Physical	Bits	Cables, hubs, radio waves

TCP/IP Model vs OSI Model



MODULE 03

Network Devices

Hubs, Switches, Routers, Firewalls & MAC Addresses

Hub vs Switch vs Router

The three core forwarding devices — each operates at a different OSI layer

HUB

Layer 1 — Physical

L1

- Broadcasts data to ALL ports
- No intelligence — doesn't know MAC/IP
- Creates a single collision domain
- Half-duplex; causes network congestion
- Largely obsolete today

SWITCH

Layer 2 — Data Link

L2

- Forwards frames using MAC address table
- Each port = separate collision domain
- Full-duplex — simultaneous send/receive
- Learns MAC addresses dynamically
- Standard in modern LANs

ROUTER

Layer 3 — Network

L3

- Routes packets between networks using IP
- Connects LAN to WAN (e.g. the Internet)
- Maintains routing tables (static/dynamic)
- Separates broadcast domains
- NAT, DHCP, firewall capabilities

More Network Devices

Essential devices that connect, protect, and extend networks

Modem

Physical / Data Link

Modulator-Demodulator

Converts digital signals to analog (and back) for transmission over telephone/cable lines. Your ISP connection starts here.

Gateway

All Layers

Protocol Translator

Connects networks using different protocols. Acts as the entry/exit point for network traffic — your router often serves as the default gateway.

Firewall

Layer 3–7

Network Security Device

Monitors and filters incoming/outgoing traffic based on security rules. Can be hardware or software. Blocks unauthorized access.

Access Point

Layer 1–2

Wireless Access Point (WAP)

Creates a wireless LAN by transmitting Wi-Fi signals. Acts as a bridge between wired and wireless networks. Identified by SSID.

NIC & MAC Addresses

NIC — Network Interface Card

Hardware component

Connects a device to a network. Built into motherboards or as a separate card.

Unique identity

Each NIC has a burned-in MAC address — assigned by the manufacturer.

Operates at Layer 2

Handles framing, error detection, and media access control (MAC).

Types

Ethernet (wired), Wi-Fi (wireless), Fiber (optical). Each has its own NIC type.

MAC Address — The Hardware ID

Example MAC Address:

A4:C3:F0	12:34:56
OUI (Manufacturer)	Device Serial

= 48-bit | 6 bytes | 12 hex digits

Globally unique

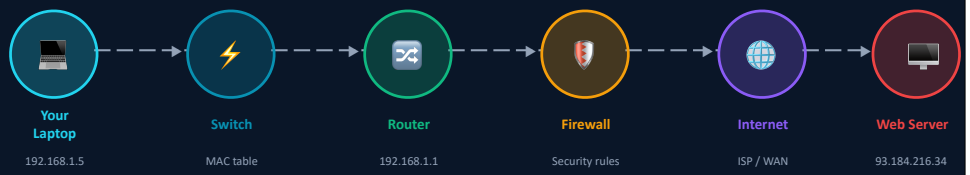
Assigned by IEEE to manufacturers. First 3 bytes = OUI (Organizationally Unique Identifier).

Used at Layer 2

Switches use MAC tables to forward frames. ARP resolves IP → MAC.

Packet Journey — Data Flow Through Devices

How a packet travels from your laptop to a web server



1 — Laptop → Switch

Frame sent using destination MAC address. Switch looks up MAC table, forwards to correct port only.

2 — Switch → Router

Packet leaves the LAN. Router inspects destination IP, selects best route via routing table.

3 — Router → Firewall → Internet

NAT translates private IP to public IP. Firewall checks rules. Packet enters the Internet via ISP.

MODULE 04

IP Addressing

IPv4 · IPv6 · Subnetting · NAT · Private vs Public

Every device on a network needs a unique address — this is how the internet routes your data.

IPv4 — Structure & Address Classes

Anatomy of an IPv4 Address: 192 . 168 . 1 . 100

192	.	168	.	1	.	100
Octet 1 (8 bits)		Octet 2 (8 bits)		Octet 3 (8 bits)		Octet 4 (8 bits)

Network Part | Host Part

IPv4 Address Classes

Cls	Range	Mask	Hosts	Use Case
A	1.0.0.0 – 126.255.255.255	/8	16M hosts	Large orgs / ISPs
B	128.0.0.0 – 191.255.255.255	/16	65K hosts	Universities / mid orgs
C	192.0.0.0 – 223.255.255.255	/24	254 hosts	Small networks (home)
D	224.0.0.0 – 239.255.255.255	–	Multicast	Streaming, routing protocols
E	240.0.0.0 – 255.255.255.255	–	Reserved	Research / experimental

Subnetting & CIDR Notation

Dividing a network into smaller sub-networks for efficiency and security

CIDR — Classless Inter-Domain Routing

192.168.1.0 / 24

The /24 is the prefix length — tells us how many bits are the network part.

Subnet Mask Bit Breakdown

/24	255.255.255.0	254 hosts
/25	255.255.255.128	126 hosts
/26	255.255.255.192	62 hosts
/30	255.255.255.252	2 hosts

Subnetting a /24 Network

192.168.1.0 / 24 → 256 addresses

Subnet A	Subnet B
192.168.1.0/26	192.168.1.64/26
.1 – .62	.65 – .126
62 hosts	62 hosts

Subnet C	Subnet D
192.168.1.128/26	192.168.1.192/26
.129 – .190	.193 – .254
62 hosts	62 hosts

IPv6 — The Next Generation

Why does IPv6 exist?

IPv4 has ~4.3 billion addresses — we ran out in 2011. IPv6 provides 340 undecillion (3.4×10^{38}) addresses — enough for every grain of sand on Earth to have trillions.

IPv6 Address Structure (128-bit, 8 groups of 16-bit hex)

2001 : 0db8 : 85a3 : 0000 : 0000 : 8a2e : 0370 : 7334

Abbreviation: Leading zeros in a group can be dropped. Consecutive all-zero groups → ::

	IPv4	IPv6
Address Size	32 bits (4 bytes)	128 bits (16 bytes)
Format	Decimal dotted (192.168.1.1)	Hex colon (2001:db8::1)
Total Addresses	~4.3 billion	~340 undecillion
Header Size	Variable (20–60 bytes)	Fixed 40 bytes
NAT Required	Yes (addresses scarce)	No — every device gets public IP
Security (IPSec)	Optional	Built-in

Public vs Private IP, NAT & Loopback

Private IP Ranges (RFC 1918)

A	10.0.0.0 — 10.255.255.255	/8 · Class A
B	172.16.0.0 — 172.31.255.255	/12 · Class B
C	192.168.0.0 — 192.168.255.255	/16 · Class C

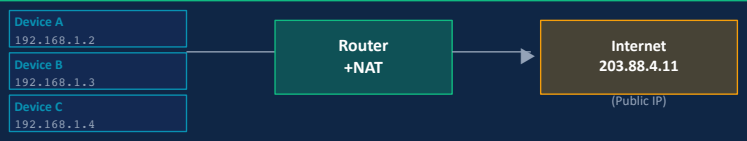
Public IP Addresses

Assigned by ISPs. Routable on the Internet.

Every home/office typically gets ONE public IP from the ISP. All internal devices share it via NAT.

`Check yours: curl ifconfig.me`

NAT — Network Address Translation



Loopback

127.0.0.1

localhost

Points back to your own machine. Never leaves the device. Used for testing local servers.

MODULE 05

TCP VS UDP

TCP: Reliability & Ordering

UDP: Speed & Low Overhead

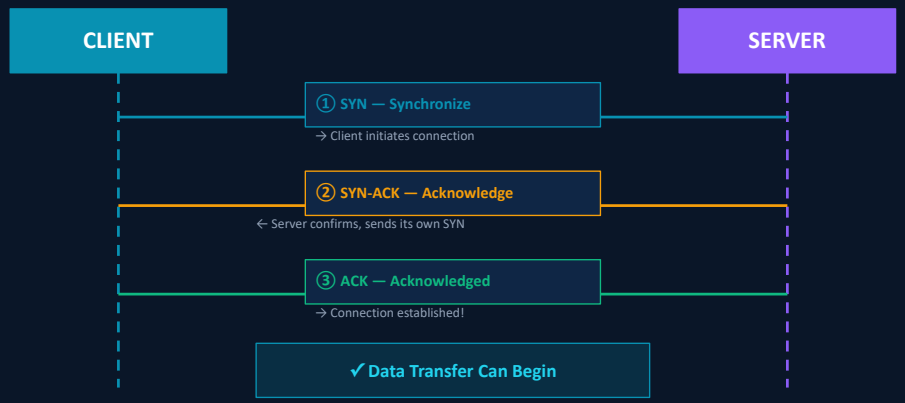
Transport Layer Protocol — The foundation of network communication

TCP vs UDP — Side by Side

TCP — Transmission Control Protocol		UDP — User Datagram Protocol
Connection Connection-Oriented (3-way handshake)		**Connection** Connectionless — fire and forget
Reliability Guaranteed delivery & ordering	VS	**Reliability** No guarantee of delivery or order
Speed Slower — overhead for reliability		**Speed** Fast — minimal overhead
Error Check Yes — acknowledgment & retransmit		**Error Check** Basic checksum only
Use Cases HTTP, FTP, Email, SSH, Databases		**Use Cases** Video streaming, DNS, VoIP, Gaming

TCP 3-Way Handshake

How a TCP connection is established before any data is sent



MODULE 06

Java Network Programming

Sockets · TCP · UDP · java.net package

java.net Package — Key Classes

Socket
`java.net.Socket`
TCP client socket. Connect to a server, get input/output streams.

ServerSocket
`java.net.ServerSocket`
TCP server socket. Listens on a port and accepts client connections.

DatagramSocket
`java.net.DatagramSocket`
UDP socket. Send and receive DatagramPacket objects.

DatagramPacket
`java.net.DatagramPacket`
Wraps UDP data with destination address and port.

InetAddress
`java.net.InetAddress`
Represents an IP address. Resolves hostnames via DNS.

URL / URLConnection
`java.net.URL`
High-level HTTP access. Open streams to web resources.

TCP Echo Server — Java Code

Server.java

```java
ServerSocket server =
  new ServerSocket(8080);

Socket client =
  server.accept(); // blocks

BufferedReader in =
  new BufferedReader(
    new InputStreamReader(
      client.getInputStream()));

PrintWriter out =
  new PrintWriter(
    client.getOutputStream(),
    true);

String msg = in.readLine();
out.println("Echo: " + msg);
client.close();
```

Socket Lifecycle

1. Create
2. Bind
3. Listen
4. Accept
5. Read/Write
6. Close

Ready to Code the Network?

Open your IDE — the internet starts with a Socket.

- ✓ Network Foundations & Types
- ✓ OSI & TCP/IP Layers
- ✓ Hands-on Lab Exercises
- ✓ TCP vs UDP Protocols
- ✓ Java Socket Programming